

SOVEREIGN

Govern Your Attention

Full Project Specification

Philosophy · Architecture · Implementation

Version 0.4, June 2026

Reclaim the right to be unreachable.

*This is not for those who hunger for connection.
It is for those who have forgotten what it feels like to be without it.*

PART I: THE MANIFESTO

Sovereign: A Manifesto for Attention

There is a moment most of us have lived but rarely name.

You are sitting across from someone you love. They are speaking. And your hand, almost without your permission, reaches for your phone.

Not because anything important happened. Not because you were bored. But because the habit is deeper than intention now. The device has become the default state. Presence has become the effort.

This is not an accident.

The Agreement You Never Signed

Somewhere in the last decade, a quiet agreement was made on your behalf. The agreement said: you will be available. Always. To everyone. Instantly.

You didn't sign it. But you accepted it, because everyone did. Because the tools that connected you to the people you love also connected you to everything else, and separating them seemed impossible.

The algorithm understood this before you did. Your attention, your most finite and most personal resource, became a commodity. Bought and sold in milliseconds. Packaged and delivered to advertisers while you scrolled, refreshed, checked, and scrolled again.

They called it connection. It was extraction.

What Sovereign Is Not

We are not anti-technology. We are not asking you to move to a cabin, abandon your WhatsApp, or explain to your mother why she can't reach you anymore.

We understand that digital infrastructure is real infrastructure. That your bank lives in an app. That your work runs on notifications. That the people you love are on platforms you didn't choose but can't leave.

We are not fighting that world. We are learning to govern our place in it.

Sovereign is not for those who hunger for connection. It is for those who have forgotten what it feels like to be without it.

The Right to Be Unreachable

| You have the right to be unreachable.

Not forever. Not from everyone. But on your terms. At your pace. By your choice.

The right to finish a meal without an interruption that wasn't urgent. The right to wake up slowly, without the world's demands already waiting on your nightstand. The right to be bored, genuinely and productively bored, and let your mind wander somewhere the algorithm can't follow.

These are not luxuries. They are the conditions of a human life.

What Sovereign Is

Sovereign is a self-hosted personal communication router. It sits between you and your digital world, evaluates every incoming message, and either delivers it to your dumb phone as a plain SMS or holds it quietly in the Vault until you choose to engage.

It is self-hosted because no company should own the infrastructure of your attention. It is open source because the community should govern what governs them. It is free because sovereignty should not require a subscription.

Sovereign. Govern your attention.

PART II: THE VOCABULARY

Every term was chosen because it reinforces the philosophy. Use these consistently across documentation, community communication, and the interface.

Term	Meaning
The Council	Your VIP list. Small by design, soft cap of 10. Their messages always reach you, skipping every other check.
The Decrees	Your rules. IF [condition] THEN [pass/hold]. Priority-ordered; lower number checked first; first match wins.
The Advisor	Local AI behavioral model (Ollama). Learns from your response patterns. Never reads message content. Never cloud inference.
The Vault	Everything held. Two tabs: Waiting and Retrieved. Nothing is ever deleted.
The Chronicle	Read-only, honest log of every Sovereign decision. Pass/hold, which gate called it, when.
The Realm	Your Sovereign instance. Your config, your data, your rules.
The Territory	Your dumb phone. The calm device that is your primary interface with the world.
The Commonwealth	The Sovereign community sharing Decrees, insights, and hardware guides.

PART III: THE ARCHITECTURE

Sovereign is a personal communication router. All seven layers are fully implemented and shipped.

```
LAYER 1 : Ingestion           - platform connectors normalize all messages
Urgency Engine               - three-tier evaluation: Council -> Decrees -> Advisor
Delivery                     - SMS formatter, quiet hours, multi-gateway support
Reply Routing                - SMS commands routed back to originating platform
Advisor                      - local Ollama behavioral model
browser-only React UI (six rooms)
LAYER 2 : The Interface      -
LAYER 3 : Deployment         - Docker Compose,
                             setup script, Makefile
LAYER 4 :
LAYER 5 :
LAYER 6 :
LAYER 7 :
```

Layer 1: Ingestion

Each connector normalizes incoming messages into a single internal schema. The urgency engine never knows which platform sent a message.

Platform	Method	Status	Invisible to sender?
WhatsApp	mautrix-whatsapp Matrix bridge	Done	Yes - they message your real number
Instagram	mautrix-meta Matrix bridge	Done	Yes - they DM your real account
Email	IMAP IDLE	Done	Yes - they email your real address
SMS	Twilio webhook	Done	Yes - they text your number
Telegram	Official Bot API (polling)	Done	No - they message your bot, not your personal account
Facebook Messenger	mautrix-meta (messenger mode)	Same bridge, not yet enabled	Yes

Important Telegram distinction: Telegram works through a bot account separate from your personal Telegram. People reach you by messaging your bot, not your real handle. The bot only sees group messages if added to the group with privacy disabled via BotFather.

Matrix Bridge Architecture (WhatsApp and Instagram)

WhatsApp and Instagram are bridged through a private Matrix homeserver (Synapse) plus mautrix bridges. This keeps the real you reachable while Sovereign filters transparently. The same Synapse homeserver and Sovereign connector handle both platforms.

```
WhatsApp <-> mautrix-whatsapp <-> Synapse (Matrix homeserver) <-> Sovereign
connectorInstagram <-> mautrix-meta <-> Synapse (Matrix homeserver) <->
Sovereign connector
```

Key behaviors:

- Portal rooms are created on demand. A chat appears only when it next receives a message. The first message in a brand-new chat may be missed; the next comes through.
- WhatsApp bridge stays linked like a WhatsApp Web companion. The phone must go online occasionally; WhatsApp unlinks idle companions after approximately 14 days.
- Instagram uses session cookies via the unofficial API. Instagram can challenge or disable the account. Enable 2FA to reduce blocks.
- Instagram contacts have no phone number. Match Council members by their bridge ID (@meta_...) rather than phone, easiest after they have messaged once.

Layer 2: The Urgency Engine

Every incoming message is evaluated through three tiers in sequence. The first tier that claims urgency wins. If none does, the message goes to the Vault.

Tier 1: The Council

The Council is checked before any Decree. A Council member's message always reaches you, including from a group, regardless of any Decree. Per-member configuration includes an optional quiet hours override toggle.

Tier 2: The Decrees

Rules are priority-ordered. Lower number is checked earlier. First match wins. Three default Decrees ship with every new Realm:

Priority	Name	Condition	Action
1	Hold all group messages	Message is in a group	HOLD - always, no override
10	Urgency keywords	emergency, urgent, hospital, accident, llamame, ayuda, please call	PASS
20	Frequency escalation	Same sender 3+ messages in 10 minutes	PASS

Because Hold all group messages sits at priority 1, a group message is held before the keyword check at 10 ever runs. A group message containing an urgency keyword still waits in the Vault. To let one specific person through a group, add them to the Council.

Tier 3: The Advisor

A local AI model that watches what you do after Sovereign makes a decision. Never reads message content. Sender names stored as scrambled codes. Runs locally via Ollama. Behavioral patterns never leave the machine. User can reset to zero at any time.

Level	Name	When	Behavior
0	Silent	Weeks 1-2	Observing only. Makes no decisions.
1	Suggesting	Weeks 3-4	Surfaces gentle suggestions in the Advisor room.
2	Advising	Month 2+	Regular recommendations with full data shown.
3	Refining	Month 4+	Auto-adjusts weights (opt-in only).
4	Sovereign	Month 6+	Fully personalized. Near-zero false signals.

Core Guardrail

Sovereign never silently drops a message. Every message either passes through or is held in the Vault. Nothing is deleted. Nothing is lost.

Layer 3: Delivery

Urgent messages are formatted as SMS and delivered to the dumb phone. Format:

```
[WA] Mama: Llamame cuando puedas, es importante[TG] Carlos: Emergency at the
office, need you now[EM] Dr. Rodriguez: Your results are ready, please call[WA]
Carlos: 4 messages in 8 min
```

SMS Gateways

Four supported gateways, auto-detected in order: twilio -> infinireach -> android -> gammu. Override with SMS_GATEWAY environment variable.

Gateway	Who it is for	Key detail
twilio	Quickest start	Set TWILIO_* credentials.
infinireach	No USB modem; hosted relay via your own SIM	INFINIREACH_API_KEY. Inbound replies require a public webhook. Local-to-local SMS avoids A2P filtering (e.g. Colombia).
android	No USB modem; fully local LAN setup	Android phone + HTTP SMS-gateway app. POST JSON {phone, message} to the send endpoint. Inbound replies via a separate SMS forwarder

		app pointing to Sovereign's /android/webhook. No public tunnel needed.
gammu	Full sovereignty	USB GSM modem or GSM/LTE HAT + GAMMU_CONFIG_PATH. SMS from your own SIM. No third party.

Note on A2P filtering: cloud long codes from Twilio and other US/EU providers are blocked in some markets (e.g. Colombia). InfiniReach and the Android gateway both use a phone's local SIM, making SMS local-to-local and avoiding this filter.

Quiet Hours

Configured in the Settings room. During quiet hours, urgent messages wait until the window ends. Two exceptions break through: Council members with their individual quiet hours override enabled, and the flood override (configurable: N urgent messages within M minutes triggers delivery regardless).

Layer 4: Reply Routing

When an urgent message reaches the dumb phone there is a 15-minute reply window. Text back a plain reply and Sovereign routes it to the correct platform and thread. Recipients see a completely normal conversation.

Command	Action
r Sounds good	Reply to the most recent urgent message
r mama I'll be home soon	Reply to the last message from contact named mama
r 2 On my way	Reply to the 2nd most recent message
list	See a numbered list of recent held messages
read 3	Read the full content of message 3
status	Get a quick vault summary

Layer 5: The Advisor

Runs locally via Ollama. Never cloud inference. Behavioral patterns never leave the machine. Sender names stored as scrambled codes, never plaintext. Message content never stored or analyzed. Reset to zero available at any time from the Advisor room with no data retained.

Layer 6: The Interface

Browser-only React UI. Password-protected (no usernames, no accounts). Changing the password signs out any other active device. Six rooms:

Room	Purpose	Key note
The Vault	Held messages: Waiting and Retrieved tabs	Opening a message moves it to Retrieved and signals the Advisor.
The Council	VIP list management	Per-member quiet hours override toggle. Soft cap reminder at 10.
The Decrees	Priority-ordered rules	Lower number = checked first. First match wins.
The Advisor	AI pattern suggestions	Maturity level, accuracy stats, reset available.
The Chronicle	Read-only decision log	Pass/hold per message, which gate called it.
Settings	Quiet hours and flood override	Global window, timezone, per-person exceptions live in Council.

Onboarding: on first load after password setup, Sovereign asks one question: who should always reach you? No feature tours, no tooltip overlays.

Layer 7: Deployment

Sovereign has one deployment model: self-hosted on hardware you physically own. There is no managed cloud tier. There is no subscription.

No company should own the infrastructure of your attention. That includes us.

Why physical hardware, not a VPS

Sovereign holds live, impersonation-grade platform session tokens (WhatsApp, Instagram, Telegram) and your full triaged message history. On rented infrastructure those bytes sit on someone else's computer, subject to their access and jurisdiction. A VPS gives software sovereignty but not data sovereignty. For this tool, that is not enough.

Hardware Options

Option	Cost	Notes
Raspberry Pi 5 (8 GB) + GSM/LTE HAT	~100 EUR one-time	Recommended appliance. Full sovereignty. SMS from your own SIM via gammu.
Old laptop / mini-PC / home server	Reuse what you own	Ideal if already available. Capable, always-on, at home.

Raspberry Pi 4 (8 GB)	~60 EUR one-time	Budget floor for the full stack. Advisor runs slower but async, never blocks routing.
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Sizing

Everything except the Advisor fits in approximately 1 GB. Ollama (the local LLM) is the heavy piece, requiring 3-4 GB while generating.

- Full stack including Advisor: Raspberry Pi 5 (8 GB) recommended. Pi 4 (8 GB) is the practical floor with a tiny model (llama3.2:1b) or Ollama offloaded to another machine via OLLAMA_URL.
- Without the Advisor: 2 GB is sufficient. Council and Decrees gatekeeping does not use the LLM.
- Always: 64-bit OS (arm64). Run from SSD/NVMe, not an SD card. Synapse and SQLite write often enough to kill SD cards.

GSM/LTE HAT vs USB modem

For a permanent Pi appliance a GSM/LTE HAT is preferable over a USB dongle: integrated, nothing to knock loose, proper antenna connector. Key considerations:

- Prefer a 4G LTE HAT (e.g. SIM7600-based) over 2G-only modules (SIM800/900). 2G and 3G networks are being shut down in many countries. Check your carrier before buying.
- Power: cellular radios draw large current bursts on transmit. Use a strong PSU (Pi 5 27W supply) and feed the HAT's dedicated power input if it has one.
- Connection: HATs present as either UART on GPIO (/dev/ttyS0 or /dev/ttyAMA0) or USB (/dev/ttyUSB*). gammu handles both.
- Docker: pass the device into the core container (devices: ["/dev/ttyUSB2:/dev/ttyUSB2"]) and point GAMMU_CONFIG_PATH at a gammu config for it.

Quick Start

```
git clone https://github.com/staysovereign/sovereign-stack.git
cd sovereign/
./setup.sh # generates secrets, checks prerequisites
# Edit .env with your values
docker compose up -d # make ollama-pull # pull the Ollama model (runs in background)
```

Open the interface at <http://localhost>. On first load, set a password, then build your Council.

With WhatsApp and Instagram

```
./setup.sh --whatsapp # prepares shared Synapse config
# Follow docs/WHATSAPP_SETUP.mddocker compose --profile whatsapp up -d
```

Common Operations

```
make logs          # tail all service logs
make status        # show running
containers and health
make backup        # dump SQLite DB to ./backups/
make update        # pull latest images and rebuild
make ollama-pull   # pull the
Ollama LLM model
make whatsapp-qr   # tail bridge logs (WhatsApp pairing /
login)
make up-whatsapp   # start full stack with WhatsApp and Instagram bridges
```

Minimum Viable Configuration

```
DUMB_PHONE_NUMBER=+1234567890
TELEGRAM_BOT_TOKEN=your-bot-token
TWILIO_ACCOUNT_SID=A
Cxxx
TWILIO_AUTH_TOKEN=xxx
TWILIO_PHONE_NUMBER=+1987654321
```

PART IV: TECHNOLOGY STACK

Component	Technology	Notes
Core engine	Python 3.12 + FastAPI	
Platform connectors	Node.js 22 + Express	
WhatsApp bridge	mautrix-whatsapp (Go, bridgev2)	Modern bridgev2 format. Config must be generated, not hand-written.
Instagram bridge	mautrix-meta (Go, Instagram mode)	Same Synapse. Appservice port 29319.
Database	SQLite (WAL mode)	Single DB for Sovereign core. Synapse uses its own SQLite.
AI / Advisor	Ollama (local LLM)	Never cloud inference. ~3-4 GB RAM while generating. Offloadable via OLLAMA_URL.
SMS gateway	Twilio / InfiniReach / Android HTTP / gammu	Auto-detected. Override with SMS_GATEWAY.
Interface	React + Vite	Password-protected. Browser-only. Six rooms.
Deployment	Docker Compose + setup.sh + Makefile	WhatsApp/Instagram via --profile whatsapp.
OS requirement	64-bit (arm64)	All images are arm64. Run from SSD/NVMe, not SD card.
License	AGPL-3.0	Anyone can run it, modify it, must keep modifications open source.

PART V: SUSTAINABILITY

Sovereign has no subscription revenue. That is intentional. The project sustains itself through models that do not create dependency or conflict of interest.

Model	What it is
Open Collective / GitHub Sponsors	Community funds development directly and transparently. Every euro visible. No strings.
Hardware bundles	Pre-configured Raspberry Pi appliance shipped ready to run. Ship the sovereignty, not the service.
Integration consulting	Paid setup and support for organizations deploying Sovereign for their teams. No access to their data.
Community documentation	staysovereign.io as the knowledge hub. The site serves users, not investors.

PART VI: LAUNCH SEQUENCE

Phase	When	Status	What
Phase 0. Founder	Now	In progress	Self as first user. Document everything: setup, friction, surprises, first week of held messages.
Phase 1. Private Beta	Month 1-2	Next	20 to 30 people from personal network. No press. No hype. Honest testers who feel the problem.
Phase 2. Community Release	Month 3-4	Planned	Repository published publicly. Manifesto published. Wiki live. staysovereign.io community opens.
Phase 3. Movement	Ongoing	Planned	Community contributes Decrees. Advisor library grows. Hardware guides multiply. Translations appear.

PART VII: KEY DECISIONS LOG

Decision	Choice Made	Rationale
Deployment model	Self-hosted on physical hardware only, no VPS, no managed cloud	Sovereign holds live platform session tokens and full message history. On rented infrastructure those bytes sit on someone else's computer. A VPS gives software sovereignty but not data sovereignty. For this tool, that is not enough.
WhatsApp connector	mautrix-whatsapp (bridgev2)	Lower ban risk for personal human-paced use. Uses modern bridgev2 format; config must be generated, not hand-written.
Instagram connector	mautrix-meta	Same Synapse reused. Risk acknowledged: unofficial API, session cookies, Instagram can challenge.
Telegram connector	Bot API, not personal account bridge	Official, stable, zero ban risk. Trade-off: contacts message the bot, not the personal account.
Database	SQLite (WAL mode)	Simpler for single-user self-hosted deployment. Easier backup via make backup.
SMS gateways	Four options: Twilio, InfiniReach, Android HTTP, gammu	Addresses A2P filtering in markets like Colombia where foreign cloud numbers are blocked.
Hardware recommendation	Raspberry Pi 5 (8 GB) + GSM/LTE HAT over VPS	Prefer 4G LTE HAT over 2G-only modules given network shutdowns. Strong PSU required for cellular radio power bursts.
AI inference	Local Ollama only, never cloud	Behavioral patterns never leave the machine. Non-negotiable.
Interface	Six rooms including Settings, browser-only, password-protected	Settings handles quiet hours and flood override. No username system.
Revenue	Open Collective + consulting + hardware bundles	No subscription means no conflict of interest between growth and user sovereignty.
Domain	staysovereign.io	Stay sovereign is a directive. Assumes the user already understands the value and is committing to it.
Repository	github.com/staysovereign/sovereign-stack	Organization account. Consistent with domain.

PART VIII: KNOWN BEHAVIORS AND LIMITATIONS

Expected behaviors by design, not bugs.

Behavior	Context	Detail
First message in new WhatsApp/Instagram chat may be missed	Matrix bridges	Portal rooms are created on demand. The connector joins a moment after the room is created. The second message comes through normally.
WhatsApp session expires after approximately 14 days of phone inactivity	mautrix-whatsapp	Bridge stays linked like WhatsApp Web. Phone must connect to the internet at least once every 14 days or the session is unlinked.
Pairing code must be entered within approximately 60 seconds	WhatsApp setup	Entering the code late triggers a known bridge refresh bug. Re-run the pairing step for a fresh code.
Group messages held even if they contain urgency keywords	Urgency engine	Hold all group messages Decree sits at priority 1, checked before keyword Decree at priority 10. Add the sender to Council to let them through from groups.
Instagram contacts have no phone number	mautrix-meta	Match Council members by bridge ID (@meta_...) rather than phone. Easiest after they have messaged once.
Telegram only sees group messages if bot is added and privacy disabled	Telegram connector	Via BotFather: /setprivacy -> your bot -> Disable.
Large Instagram first sync creates many portals	mautrix-meta	Connector paces its joins to stay under Synapse rate limits. Expected during initial setup.
Email startup backlog behavior	Email connector	EMAIL_STARTUP_SCAN_LIMIT controls how many pre-existing unseen messages are processed on boot. Default 10. Set to 0 to skip backlog. Mail arriving after startup is always processed live.
Android gateway is send-only by default	Android SMS gateway	Most gateway apps cannot forward SMS the SIM receives. Pair with a separate incoming-SMS forwarder app pointing to Sovereign's /android/webhook on the LAN. No public tunnel needed.
LLM RAM requirement	The Advisor / Ollama	Ollama requires approximately 3-4 GB RAM while generating. Pi 4 (4 GB) users should use llama3.2:1b or offload Ollama to another machine via OLLAMA_URL.
SD card degradation	Raspberry Pi deployment	Synapse and SQLite write frequently enough to kill SD cards. Always run from SSD or NVMe.

PART IX: COMMUNITY AND DOCUMENTATION

Resource	URL	What it contains
Landing page	staysovereign.io	Project home, philosophy, how it works
GitHub repository	github.com/staysovereign/sovereign-stack	Source code, README, setup scripts
Wiki	github.com/staysovereign/sovereign-stack/wiki	Full documentation: installation, platform setup, Decrees guide, troubleshooting
Discussions	github.com/staysovereign/sovereign-stack/discussions	The Commonwealth: setups, Decree sharing, questions, philosophy

CLOSING NOTE

Version 0.4 reflects a significant maturation of the deployment philosophy. The project moved from 'self-hosted recommended' to 'self-hosted on physical hardware only', grounded in a precise argument: Sovereign holds live platform session tokens and your full message history. On rented infrastructure those bytes sit on someone else's computer. A VPS gives software sovereignty but not data sovereignty. For this tool, that is not enough.

Every other technical decision traces back to the same root. The Raspberry Pi 5 plus a GSM/LTE HAT is not just a cheap server. It is the most complete expression of the philosophy achievable with consumer hardware. Someone receives a device, plugs it in at home, and is sovereign from that moment.

All seven architectural layers are complete. The Wiki and Discussions are live. The next chapter is Phase 1: putting it in the hands of 20 to 30 real people who felt that feeling.

This is not for those who hunger for connection. It is for those who have forgotten what it feels like to be without it.

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staysovereign.io · github.com/staysovereign/sovereign-stack